

Urvashi Gondaliya | Research Scientist

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PROFESSIONAL SUMMARY

Dedicated **Research Professional** with **4.5+ years of experience** in materials science research across academic and industrial environments. Known for improving experimental workflows, enhancing material performance, and reducing testing errors through process optimization. Experienced in leading end-to-end research activities, publishing peer-reviewed work, and supporting lab-scale to pilot-scale transitions. Proven ability to maintain safety standards, ensure data accuracy, and contribute to impactful research outcomes across diverse material applications.

TECHNICAL SKILLS

Electrochemical Testing: HER, OER, UOR, MOR, ORR, CV, GCD, EIS, ECSA, Chronoamperometry
Material Characterization: XRD, SEM, FTIR, TGA, DMA, DSC, Tensile Testing
Synthesis Methods: Hydrothermal synthesis, Microwave-assisted synthesis, Sonication, Tube furnace, Electrode coating
Fabrication Techniques: 3D printing, Injection molding, Polymer feedstock optimization, Composite sample processing
Data Analysis Tools: Origin Pro, MS Excel (advanced charting, statistical analysis)
Lab Safety & Quality: Lab safety compliance, Quality control, Equipment handling, Test protocol design
Scientific Communication: Manuscript preparation, Research documentation, Peer-reviewed publishing, Proposal data support
Software: Origin Pro, Astrometrica PS1 & PS2, Versa Studio, MS Excel.

WORK EXPERIENCE

National Institute of Materials Advancement (NIMA), KS, USA Research Assistant	Aug 2023 – Current
- Tested over 30 catalyst samples across HER, OER, UOR, MOR, and ORR reactions, increasing energy conversion efficiency by 18% using electrochemical evaluation methods with precise control parameters. - Assessed 20+ electrode materials for supercapacitor applications using GCD, CV, EIS, ECSA, and chronoamperometry, improving validation speed by 25% and enabling quicker feedback for synthesis optimization. - Characterized 50+ nanomaterial samples using XRD and SEM, improving analysis accuracy by 30% while supporting morphology evaluation for catalytic property enhancement across varied synthesis conditions. - Executed over 100 laboratory tests while enforcing strict safety compliance, maintaining zero incidents, and preserving uninterrupted research workflows across multiple catalyst performance studies. - Composed three peer-reviewed manuscripts, converting raw experimental data into publishable results that supported funding proposals exceeding \$200K and increased lab output visibility within academic networks. - Reduced experimental reruns by 22% by identifying inconsistencies in reaction outcomes and adjusting electrolyte concentrations, scan rates, and temperature parameters during live catalyst tests.	
Reliance Industries, GJ, India Research Scientist	Feb 2021 – Jul 2023
- Synthesized nanomaterials via hydrothermal and microwave methods to develop advanced coatings, improving heat resistance and performance of polymer-based materials by over 15%. - Characterized material structure and morphology using XRD, SEM, FTIR, and TGA, enabling accurate validation of thermal stability and crystalline behavior in polymer formulations. - Operated tube furnace and sonicator for controlled processing of composite samples, ensuring consistent synthesis quality during lab-to-pilot scale transitions. - Designed mechanical testing protocols using tensile, DMA, and DSC instruments to evaluate material strength, achieving 12% improved durability in polymer prototypes. - Utilized Origin Pro and Excel for data plotting, statistical analysis, and experimental visualization, reducing result reporting time across R&D workflows by 30%. - Fabricated components using 3D printing and injection molding techniques; optimized polymer feedstock blends to achieve precise dimensional accuracy and structural consistency.	

EDUCATION

Master of Science in Material Science Pittsburg State University, KS, USA	May 2025
Master of Science in Physics Gujarat University, India	Apr 2022
Bachelor of Science in Physics Gujarat University, India	Sep 2020

CONFERENCES

- Targeted synthesis of Unique Nickel Sulfide (NiS, NiS₂) Microarchitectures and the application for the Enhanced Water Splitting System (NIMA-ICEMT-2024 Pittsburg)
- Metal Disulfide Fe/CoS/Ni as electrocatalyst for overall water splitting (ACS-MWRM-2024 Omaha)
- Mesoporous Hollow Nested Nanospheres of Ni, Cu Co-Based Mixed Sulfides for Electrocatalytic Oxygen Reduction and Evolution (NIMA-ICEMT-2025 Pittsburg)
- Synthesis of nanostructured Metal Transition Metal Sulfides for electrocatalytic Application (ACS-SWRM-2024 San Diego)

PROJECTS

- Pittsburg State University | Mar 2023 – May 2024**
- Investigated metal disulfides (Fe, Co, Ni) electrocatalysts for efficient overall water splitting, enhancing catalytic activity in hydrogen and oxygen evolution reactions.
 - Analyzed phase transitions of nickel sulfide nanoparticles and correlated structural changes to improved catalytic performance in hydrogen evolution (HER) and oxygen evolution reactions (OER).

AWARDS & ACHIEVEMENTS

- Participated in the Asteroid Search Campaign
- Attended the ACS Regional Conference, Omaha, NE, October 2024